***p. 68* There is a civilization 200 light-years away, in the constellation of Andromeda.**

A for Andromeda and its sequel, *Andromeda Breakthrough*, are inconsistent about whether the alien civilization hails from the enormously distant Andromeda *galaxy*, or a nearer star in the constellation of Andromeda as I said. In the first novel the planet is placed 200 light-years away, well within our own galaxy. In the sequel, however, the same aliens are located in the Andromeda galaxy, which is about 2 million light-years away. Readers of my page 68 may replace '200' with '2 million' according to taste. For my purpose the relevance of the story remains undiminished.

Fred Hoyle, the senior author of both these novels, is an eminent astronomer and the author of my favourite of all science fiction stories, *The Black Cloud*. The superb scientific insight deployed in his novels makes a poignant contrast to his spate of more recent books written jointly with C. Wickramasinghe. Their misrepresenting of Darwinism (as a theory of pure chance) and their waspish attacks on Darwin himself do nothing to assist their otherwise intriguing (though implausible) speculations on interstellar origins of life. Publishers should correct the misapprehension that a scholar's distinction in one field implies authority in another. And as long as that misapprehension exists, distinguished scholars should resist the temptation to abuse it.

***p. 71* … strategies and tricks of the living trade …**

This strategic way of talking about an animal or plant, or a gene, as if it were consciously working out how best to increase its success—for instance picturing 'males as high-stake high-risk gamblers, and females as safe investors' (p. 73)—has become commonplace among working biologists. It is a language of convenience which is harmless unless it happens to fall into the hands of those ill-equipped to understand it. Or over-equipped to misunderstand it? I can, for example, find no other way to make sense of an article criticizing *The Selfish Gene* in the journal *Philosophy*, by someone called Mary Midgley, which is typified by its first sentence: 'Genes cannot be selfish or unselfish, any more than atoms can be jealous, elephants abstract or biscuits teleological.' My own 'In Defence of Selfish Genes', in a subsequent issue of the same journal, is a full reply to this incidentally highly intemperate and vicious paper. It seems that some people, educationally over-endowed with the tools of philosophy, cannot resist poking in their scholarly apparatus where it isn't helpful. I am reminded of P. B. Medawar's remark about the attractions of 'philosophy-fiction' to 'a large population of people, often with well-developed literary and scholarly tastes, who have been educated far beyond their capacity to undertake analytical thought'.

***p. 76* Perhaps consciousness arises when the brain's simulation of the world becomes so complete that it must include a model of itself.**

I discuss the idea of brains simulating worlds in my 1988 Gifford Lecture, 'Worlds in Microcosm'. I am still unclear whether it really can help us much with the deep problem of consciousness itself, but I confess to being pleased that it caught the attention of Sir Karl Popper in his Darwin Lecture. The philosopher Daniel Dennett has offered a theory of consciousness that takes the metaphor of computer simulation further. To understand his theory, we have to grasp two technical ideas from the world of computers: the idea of a virtual machine, and the distinction between serial and parallel processors. I'll have to get the explanation of these out of the way first.

A computer is a real machine, hardware in a box. But at any particular time it is running a program that makes it look like another machine, a virtual machine. This has long been true of all computers, but modern 'user-friendly' computers bring home the point especially vividly. At the time of writing, the market leader in user-friendliness is widely agreed to be the Apple Macintosh. Its success is due to a wired-in suite of programs that make the real hardware machine—whose mechanisms are, as with any computer, forbiddingly complicated and not very